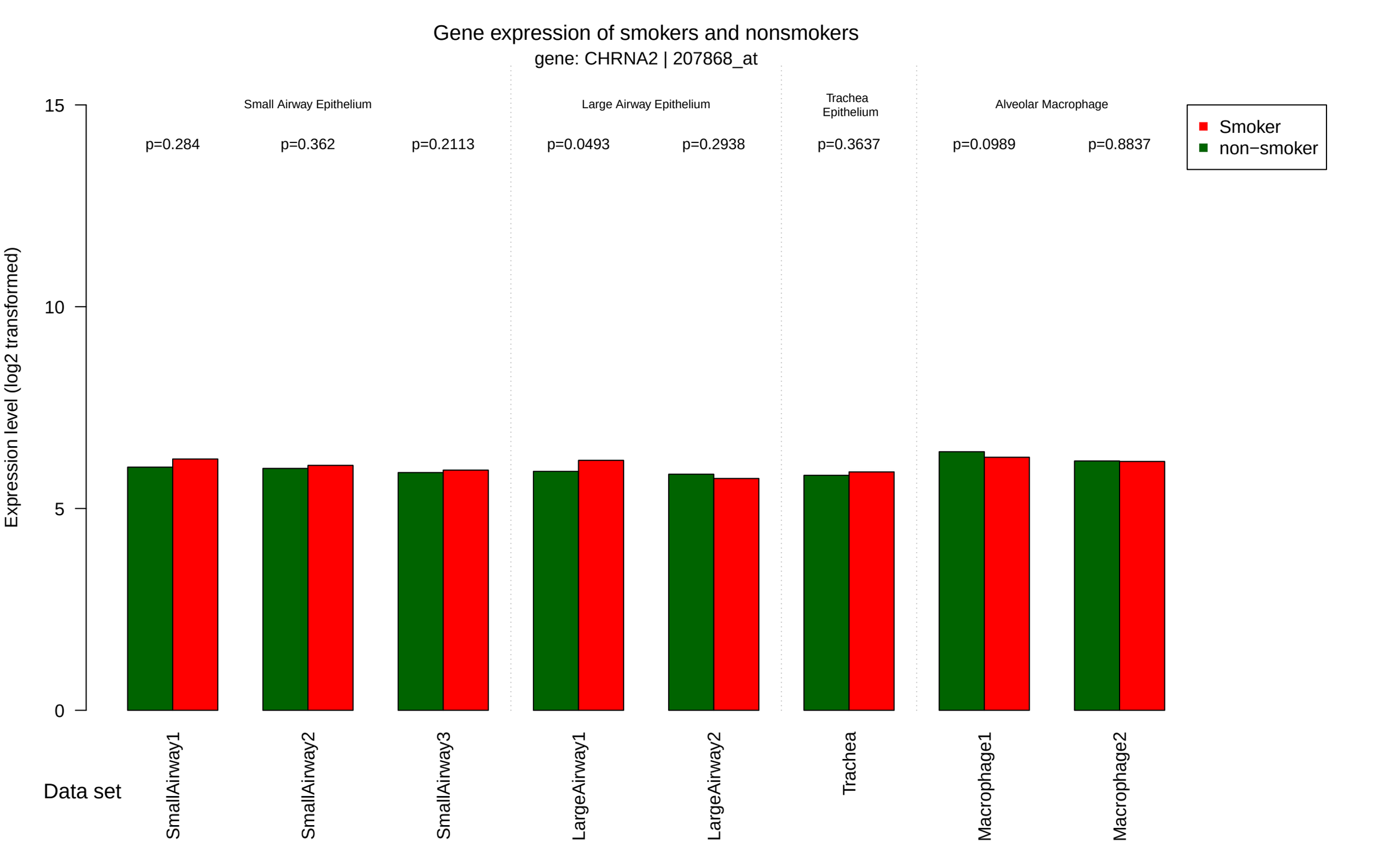
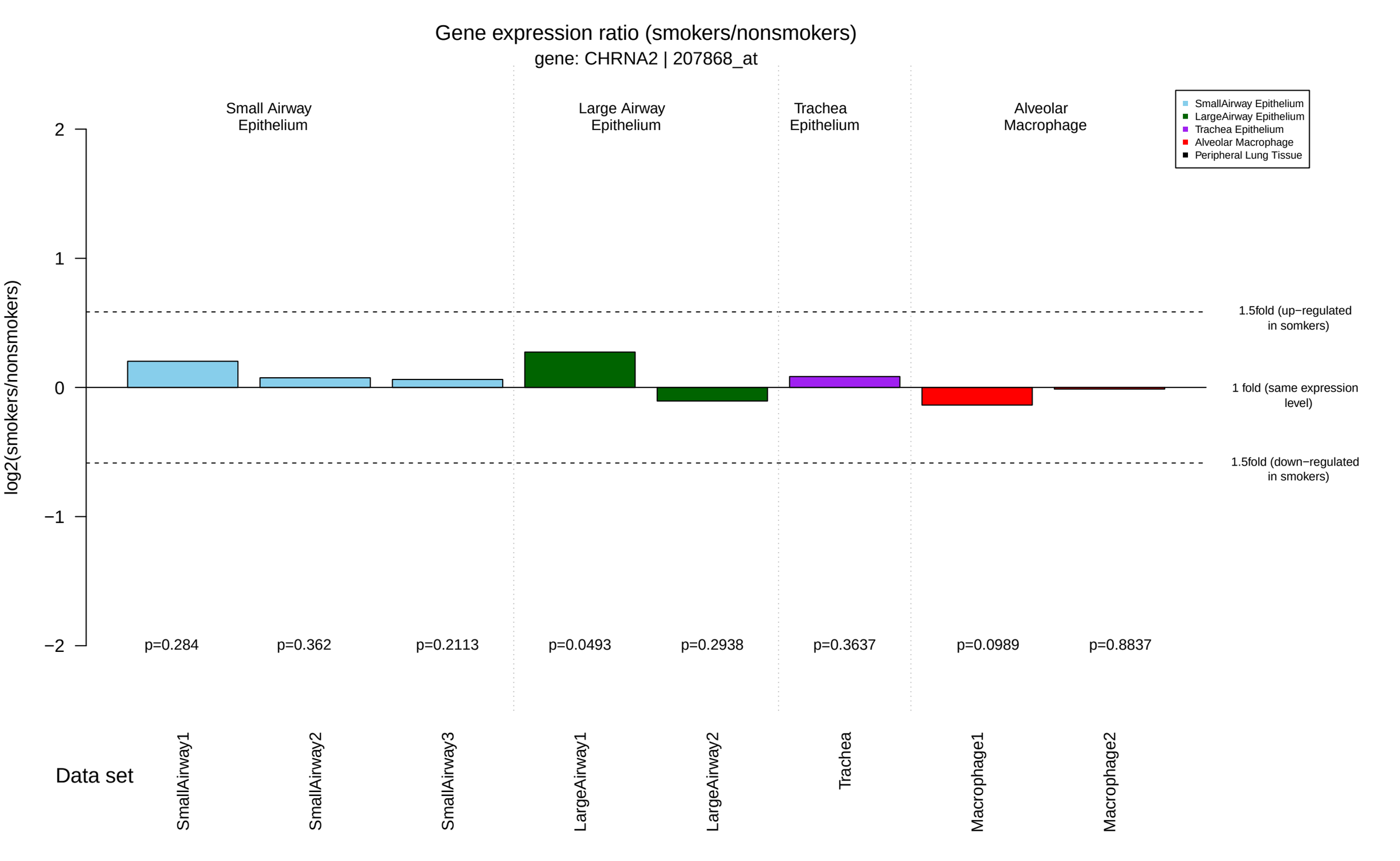
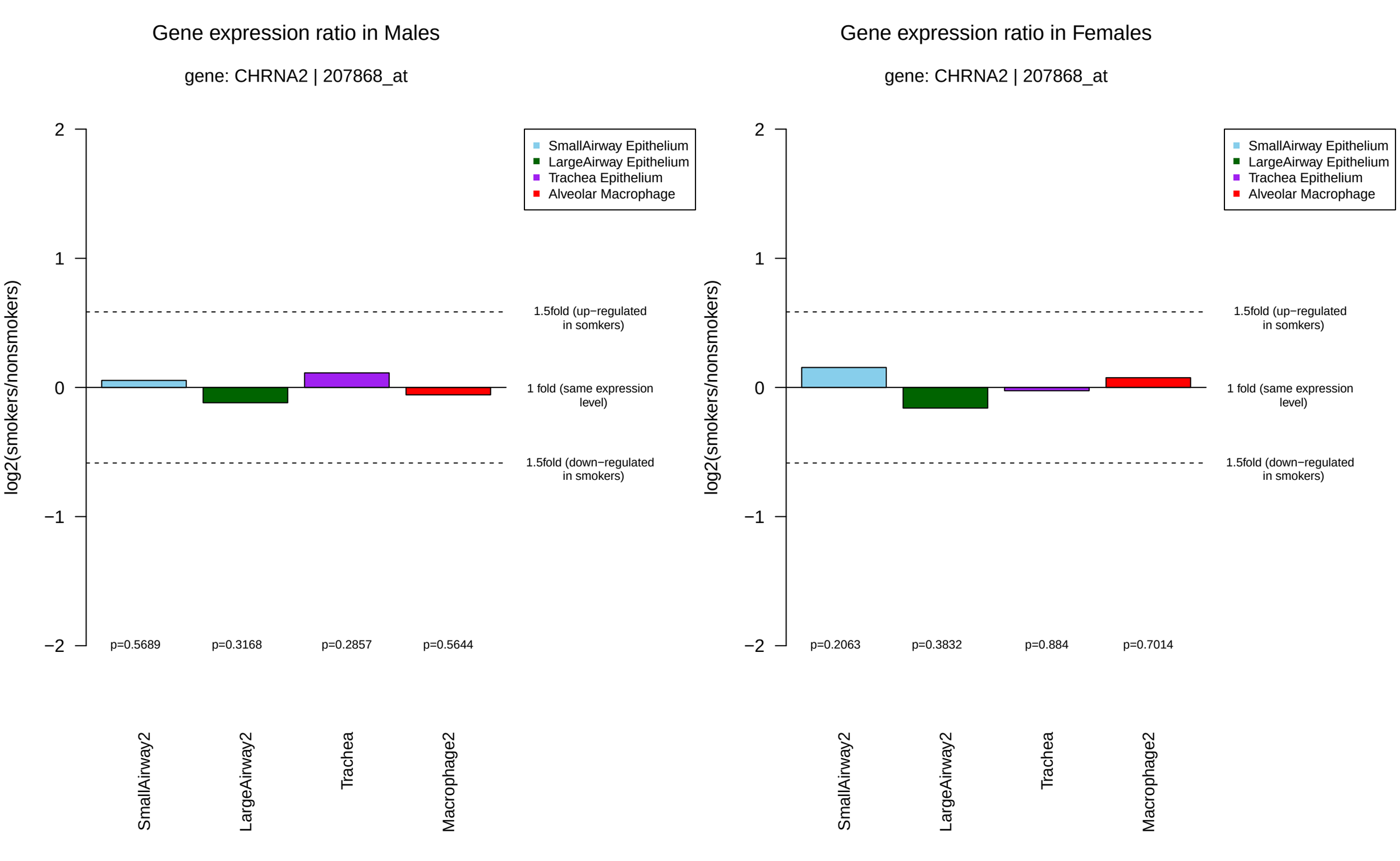
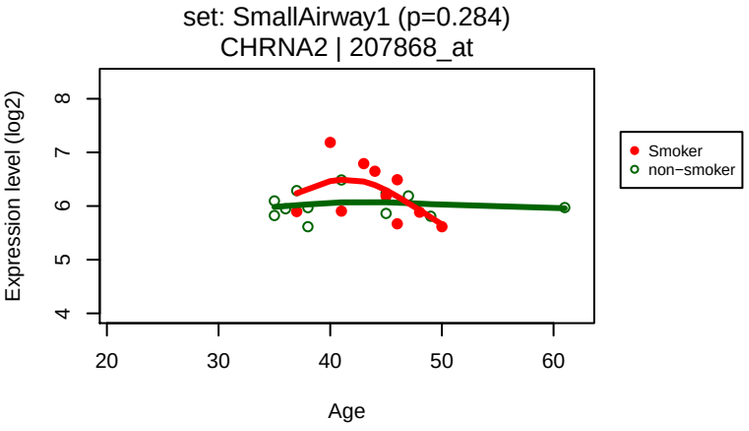
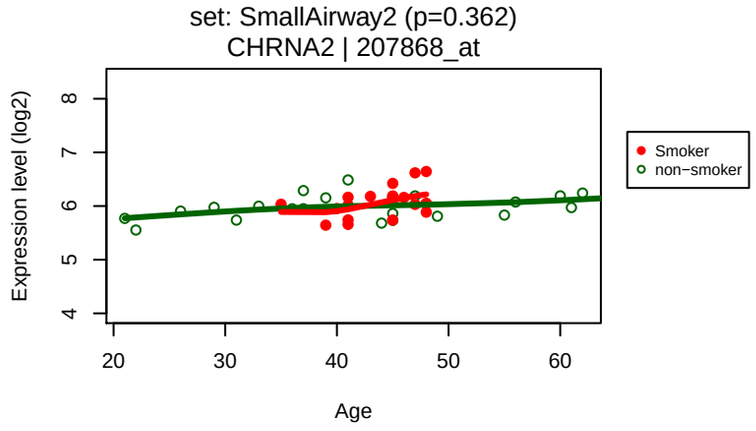
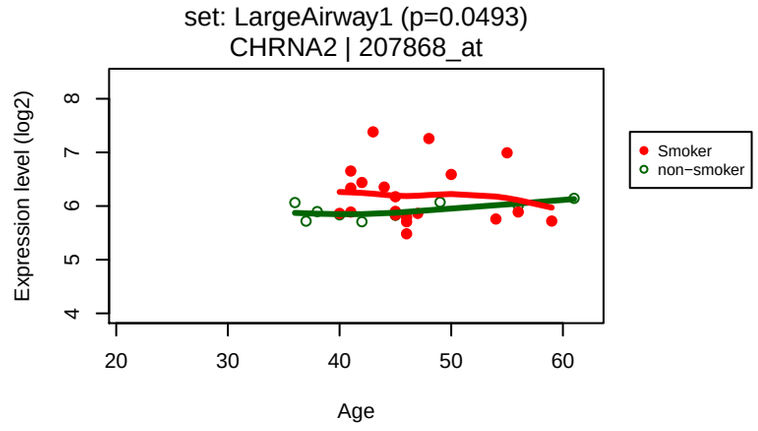

Gene expression of smokers and nonsmokers
gene: CHRNA2 | 207868_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.284
p=0.362
p=0.2113
p=0.0493
p=0.2938
p=0.3637
p=0.0989
p=0.8837
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

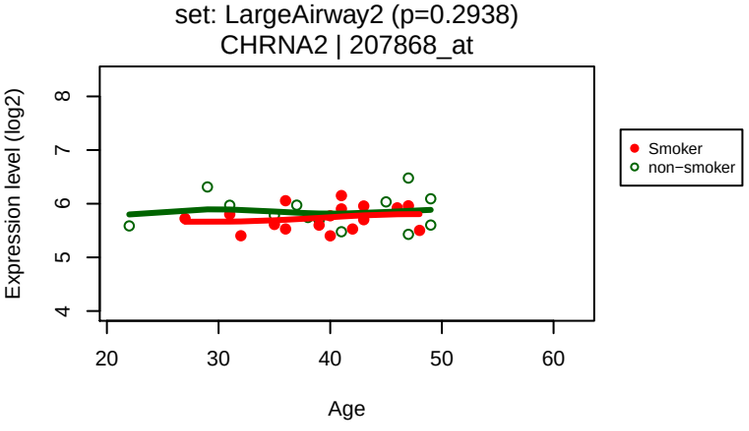

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA2 | 207868_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.284
p=0.362
p=0.2113
p=0.0493
p=0.2938
p=0.3637
p=0.0989
p=0.8837
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

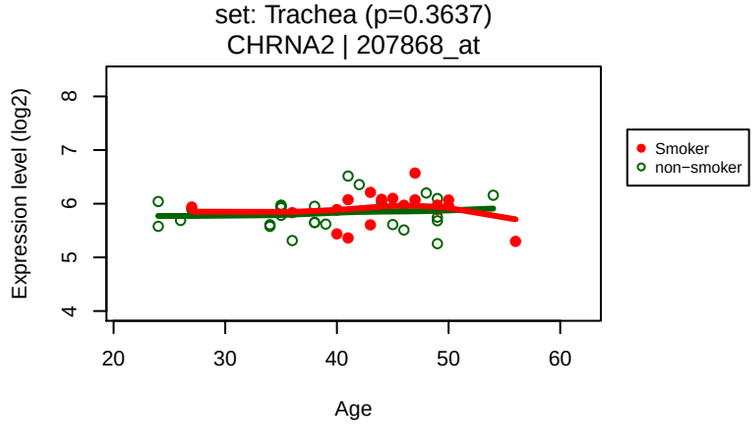

Gene expression ratio in Males
gene: CHRNA2 | 207868_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5689
p=0.3168
p=0.2857
p=0.5644
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA2 | 207868_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2063
p=0.3832
p=0.884
p=0.7014
SmallAirway2
LargeAirway2
Trachea
Macrophage2

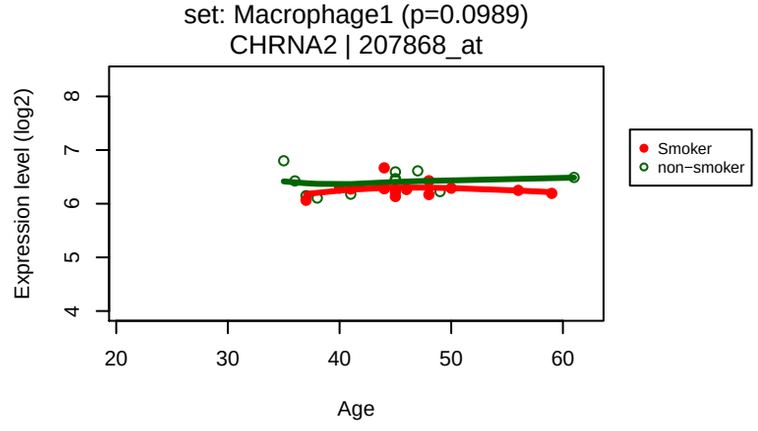

set: SmallAirway1 (p=0.284)
CHRNA2 | 207868_at
Expression level (log2)
Age
Smoker
non−smoker

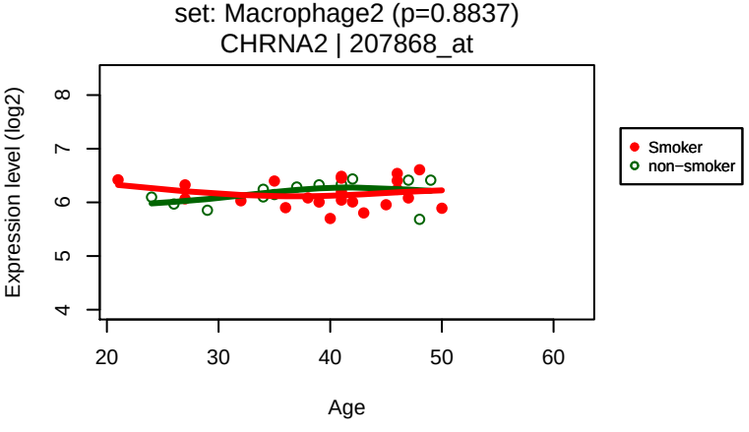

set: SmallAirway2 (p=0.362)
CHRNA2 | 207868_at
Expression level (log2)
Age
Smoker
non−smoker

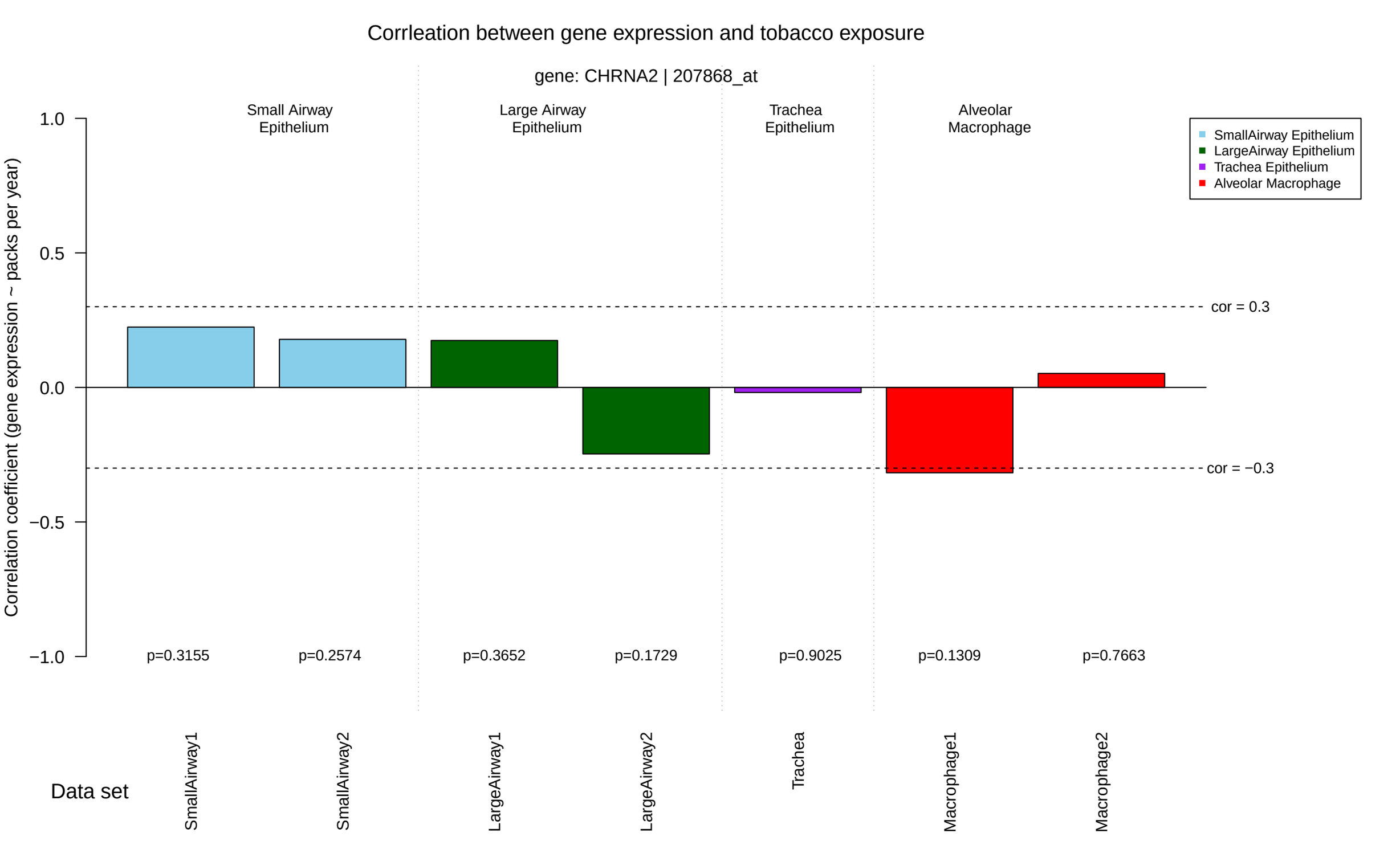

set: LargeAirway1 (p=0.0493)
CHRNA2 | 207868_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2938)
CHRNA2 | 207868_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3637)
CHRNA2 | 207868_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0989)
CHRNA2 | 207868_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8837)
CHRNA2 | 207868_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA2 | 207868_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3155
p=0.2574
p=0.3652
p=0.1729
p=0.9025
p=0.1309
p=0.7663
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2